

$$D(s) = s^3 + 3s^2 + 2s + K^*$$



$$D(j\omega) = -j\omega^3 - 3\omega^2 + j2\omega + K^*$$



$$\operatorname{Re}D(j\omega) = -3\omega^2 + K^*, \operatorname{Im}D(j\omega) = -\omega^3 + 2\omega$$



$$\omega = \pm\sqrt{2}, K^* = 6$$



imaginary-axis crossing: $s = \pm j\sqrt{2}$